DRAFT FOR REVIEW, OCTOBER 2026

Absorbing an acquired bank's tech stack

Workflow design for POCs 1, 2, 5 and 6

The unified design stays as it is. This deck goes into four of its modules and shows their Orkes workflows task by task. A merger is the example because it needs all four.

POC 2 compares the two stacks and proposes a migration plan. POC 1 turns that plan into a workflow and runs it wave by wave, with approval at each wave. POC 5 repairs integrations that break during cutover, and POC 6 checks that the old and new systems agree.

Each workflow still runs on its own. The merger is one way of chaining them, and it is set up through configuration.

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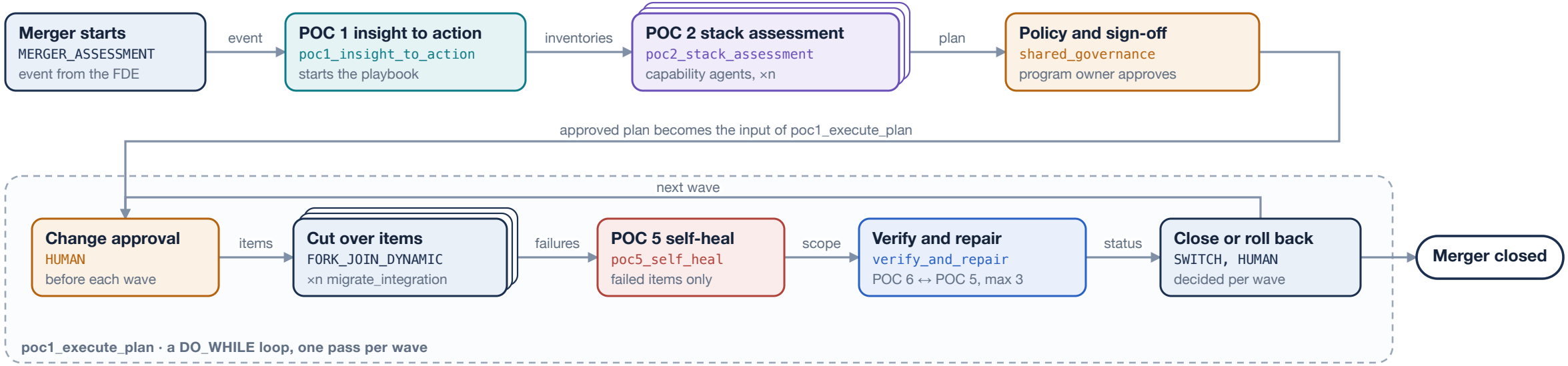
THE MERGER, STEP BY STEP

1	**Assess** Compare both stacks and recommend keep, replace or migrate for each system.	POC 2
2	**Plan** Order the moves into waves by dependency. Architects review the risky ones.	POC 2
3	**Cut over** Run each wave once the change board approves it.	POC 1
4	**Heal** Repair integrations that break on the new systems.	POC 5
5	**Reconcile** Compare old and new records. Fix the safe gaps and raise the rest.	POC 6
6	**Close or roll back** Close the wave, or move its integrations back to the old route.	POC 1

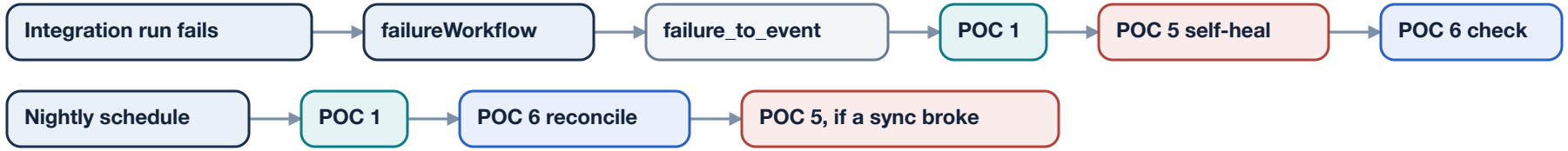
END TO END

How the four workflows hand work to each other

PLAN



AFTER GO-LIVE



Each arrow is one workflow's output used as the next one's input. In Orkes that is a SUB_WORKFLOW whose output a later task reads, for example `${assess_ref.output.recommendedAction}` passed into poc1_execute_plan. After go-live, failures and a nightly schedule start the same workflows.

BUILDING BLOCKS

Twelve workflows that share one input and output format

The contract

INPUT

```
{
  "eventType": "RECON_REQUEST",
  "correlationId": "mrg-042",
  "context": { "waveId": 2 },
  "payload": {
    "entities": ["accounts", "balances"],
    "systems": ["bank_b_core", "bank_a_core"]
  }
}
```

OUTPUT

```
{
  "status": "NEEDS_REPAIR",
  "recommendedAction": {
    "type": "REPAIR_INTEGRATION",
    "integrationId": "billpay"
  },
  "risk": "MEDIUM",
  "evidence": ["12 transactions missing in bank_a_core"],
  "correlationId": "mrg-042"
}
```

Every run for one merger carries the same correlation ID, using Orkes' own correlationId field, so all of a merger's runs can be found together. Wave and item IDs travel in `context`.

The workflows

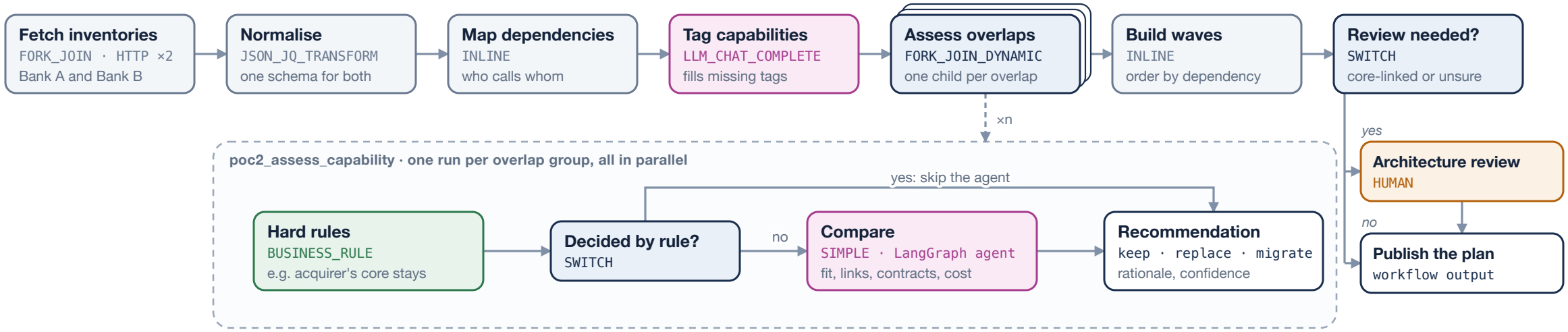
WORKFLOW	OWNER	STARTED BY	RETURNS
`poc1_insight_to_action`	POC 1	insights from POCs, events, failure adapter, schedule	the outcome of the action workflow
`poc1_execute_plan`	POC 1	poc1_insight_to_action	each wave closed, or rolled back
`shared_governance`	Shared	POC 1 workflows	auto, approved or rejected
`verify_and_repair`	Shared	POC 1 workflows	matched, or exceptions after 3 rounds
`poc2_stack_assessment`	POC 2	poc1_insight_to_action	migration plan in waves
`poc2_assess_capability`	POC 2	dynamic fork in POC 2	keep, replace or migrate, with rationale
`poc5_self_heal`	POC 5	POC 1 workflows	recovered, fallback or escalated
`poc6_reconcile`	POC 6	POC 1 workflows	matched, fixed, needs repair or exceptions
`poc6_compare_batch`	POC 6	dynamic fork in POC 6	mismatches and control totals
`migrate_integration`	Adapter	dynamic fork in poc1_execute_plan	moved, or failed
`rollback_wave`	Adapter	poc1_execute_plan	old routes restored
`failure_to_event`	Adapter	failureWorkflow of any integration run	starts poc1_insight_to_action

POC 2 ASSESS AND PLAN

Build: week 1

Stack assessment workflow

`poc2_stack_assessment` In: both banks' application and integration inventories. Out: a migration plan in waves, with the evidence for each move.



Rules · AI · Human · Call or transform · Flow control · Another POC (its colour)

HUMAN GATE

Architecture review when an item touches a core system or the agent's confidence is under the threshold. Reviewers approve, edit or reject each item.

AN FDE COACH CONFIGURES

Field mappings for each bank, the capability list, the rules sheet, scoring weights, the review threshold and the wave size.

DEMO

Both mock banks run a loan origination system. The agent recommends keeping Bank A's and moving Bank B's loans in a later wave. Because Bank B's system feeds the core, an architect has to confirm.

BRING YOUR OWN AGENT

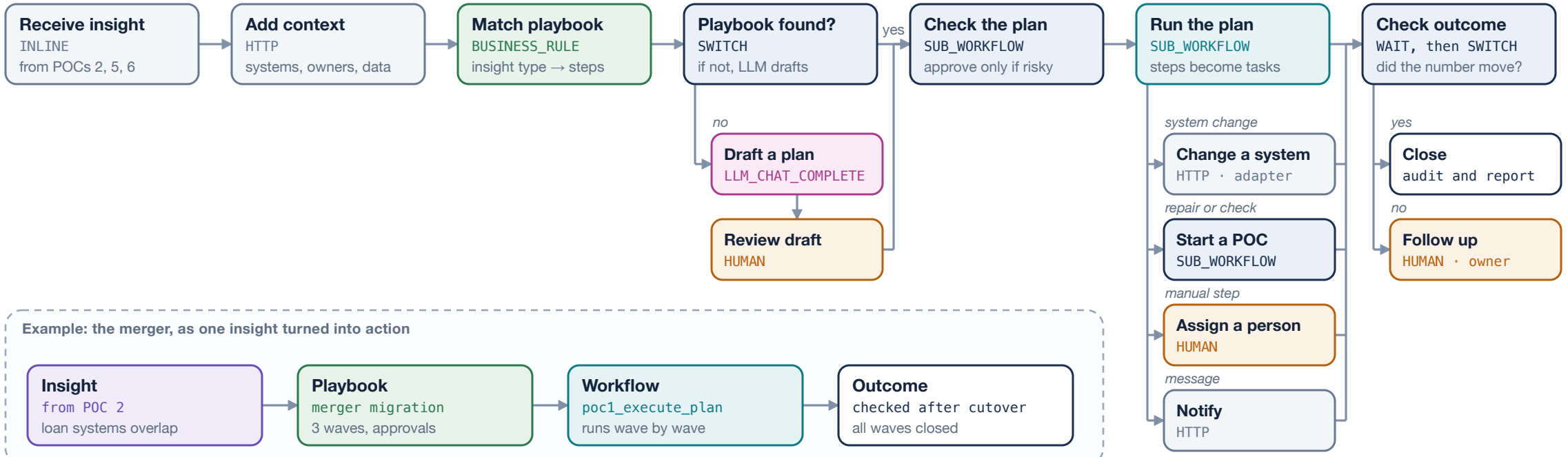
Compare is a LangGraph agent running as an Orkes worker, to show that Orkes can run agents built in other frameworks. An Agno agent could take the same slot.

POC 1 INSIGHT TO ACTION

Build: week 1

Intelligence-to-action workflow

`poc1_insight_to_action` In: an insight from POC 2, 5 or 6, or from analytics. Out: a workflow that acts on it, and a check that it worked.



Rules · AI · Human · Call or transform · Flow control · Another POC (its colour)

HUMAN GATES

The policy sheet decides when a plan needs approval. A plan drafted by the LLM always goes to a person before it runs.

AN FDE COACH CONFIGURES

Playbooks (insight type to steps), the policy sheet, approver groups, and how long to wait before checking the outcome.

DEMO

POC 2 reports that the two banks run overlapping loan systems. POC 1 matches this to the merger playbook and turns it into a three-wave plan. Once approved, the plan runs as poc1_execute_plan.

CLOSING THE LOOP

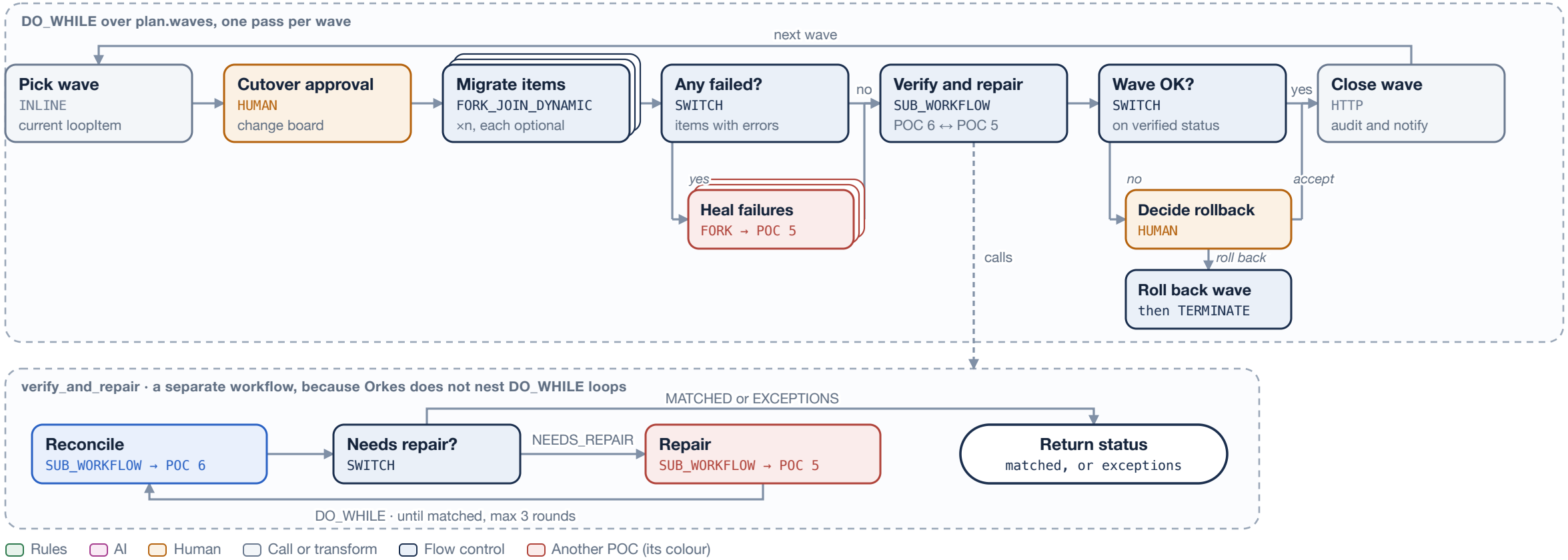
Each plan names the number it should change, such as failed calls or open mismatches. POC 1 checks it after the wait and reopens the case if it has not moved.

POC 1 CUT OVER

Build: week 2

Running the migration plan, one wave at a time

`poc1_execute_plan` In: the approved plan. Out: every wave closed, or rolled back with the reason recorded.



WHY WAVES

A failed wave touches only its own integrations, so a rollback has a clear boundary.

HUMAN GATES

The change board before every wave. A rollback decision when the data still disagrees after the last repair round.

AN FDE COACH CONFIGURES

Wave order and size (from POC 2), cutover windows, the change board, repair rounds and whether fallback is allowed.

DEMO

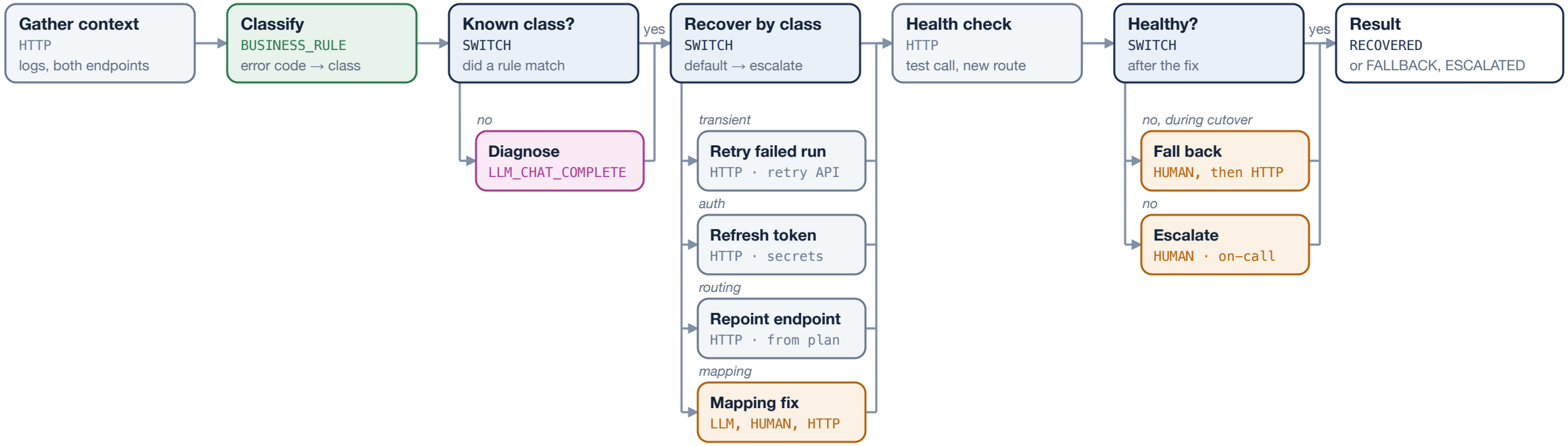
Wave 1 closes clean. In wave 2, POC 5 and POC 6 clear 401 errors on bill-pay, a field-format error and twelve missing transactions.

POC 5 HEAL

Build: week 2

Self-healing workflow for cutover failures

`poc5_self_heal` In: failure details from POC 1. Out: recovered, moved back to the old route, or escalated with a diagnosis.



Rules AI Human Call or transform Flow control Another POC (its colour)

TYPICAL MERGER FAILURES

New endpoints not live yet, credentials not provisioned, field formats that differ between the two cores, rate limits on the new system.

HUMAN GATES

Mapping changes and moving traffic back to the old route need approval. Retries, a token refresh and repointing to a planned endpoint do not.

AN FDE COACH CONFIGURES

The error-code rules sheet, retry limits, the list of safe fixes, whether fallback is allowed during cutover, and the on-call group.

DEMO

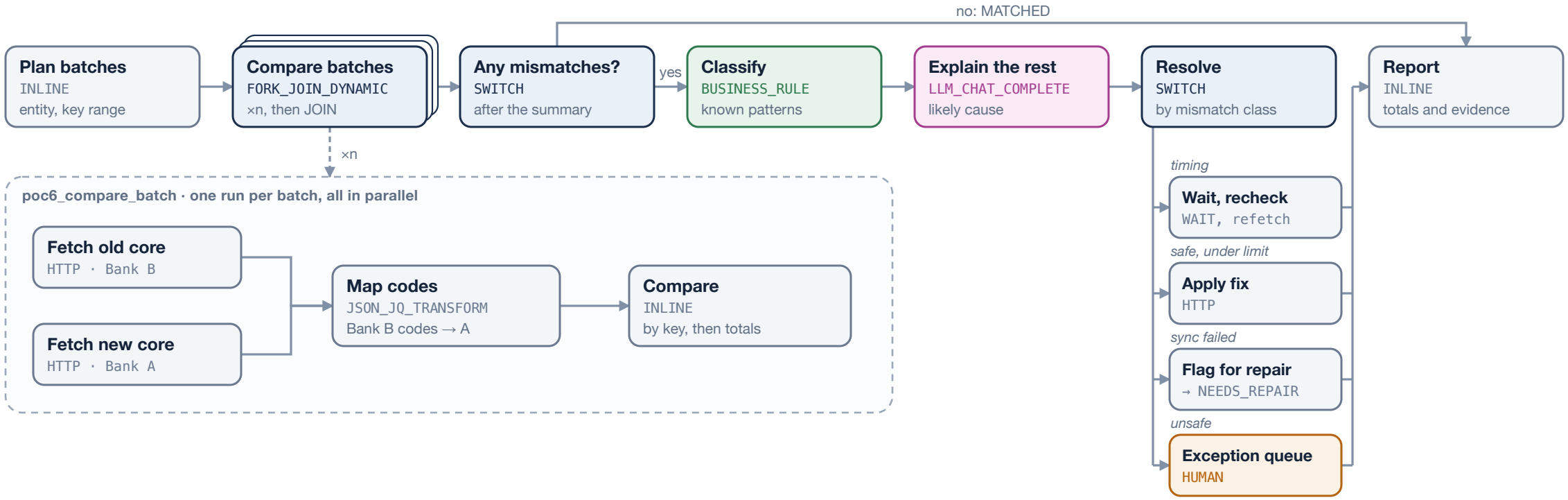
After cutover, Bank B's bill-pay integration fails with 401 errors. POC 5 refreshes the token from Orkes secrets and retries the failed run through the retry API. The health check passes.

POC 6 RECONCILE

Build: week 3

Reconciliation workflow

`poc6_reconcile` In: a scope (entities, systems, as-of time). Out: matched, fixed, needs repair, or exceptions, with control totals.



Rules · AI · Human · Call or transform · Flow control · Another POC (its colour)

WHAT IT COMPARES

Record counts, balances and key fields in Bank B's old core and Bank A's core, after code mapping so the two sides line up.

HUMAN GATE

Mismatches over the auto-fix limit, and any cause the rules and the LLM cannot explain, go to the exception queue.

AN FDE COACH CONFIGURES

Entities and fields, matching keys, code mappings, tolerances, the auto-fix limit, batch size and the recon team group.

DEMO

After wave 2, twelve transactions are missing in Bank A's core. POC 6 traces them to the bill-pay outage and returns NEEDS_REPAIR. POC 5 replays them and the second round matches.

WHEN THINGS GO WRONG

What happens when something fails

WHAT GOES WRONG	CAUGHT BY	WHAT HAPPENS NEXT
A call fails while an integration is moved	migrate_integration (task retries, marked optional)	POC 1 hands the item to POC 5 before the wave is verified.
POC 5 cannot fix it during cutover	`poc5_self_heal`	With approval, traffic goes back to the old route. Otherwise on-call gets the diagnosis.
Data still disagrees after three rounds	`verify_and_repair`	POC 1 asks the change board whether to roll the wave back or accept the exceptions.
A wave is rolled back	`rollback_wave`	Its integrations return to Bank B's routes. The audit log records who decided and why.
Nobody approves in time	HUMAN escalation chain	The task moves to the next approver. The wave waits for its next window.
A workflow itself fails	failureWorkflow, then failure_to_event	The failure becomes an ordinary event for POC 1, which hands it to POC 5 or to a person.
The LLM is unsure	Any LLM step	A low-confidence answer never acts alone. It goes to a person with the evidence.

> Nothing changes a bank system unless a policy rule allows it or a named person approved it.

REUSE

For a new merger, only the configuration changes

merger.yaml (example values)

```yaml
merger:
  id: mrg-042
  acquirer: bank_a        # endpoints: ${workflow.env.bank_a}
  target: bank_b          # keys: ${workflow.secrets.bank_b}

assessment:
  inventories: [bank_a_apps.csv, bank_b_apps.csv]
  rules: merger_rules.xlsx   # BUSINESS_RULE sheet
  reviewBelowConfidence: 0.7
  waveSize: 4

cutover:
  approvers: change-board
  window: "Sat 01:00-05:00"
  maxRepairRounds: 3
  allowFallback: true

reconcile:
  entities: [accounts, balances, customers]
  amountTolerance: 0.00
  autoFixLimit: 250
  exceptionsTo: recon-team
```

Where each setting lives in Orkes

SETTING	LIVES IN
Endpoints for both banks	Orkes environment variables
Credentials	Orkes secrets
Playbooks, merger rules, policy, error classes	BUSINESS_RULE spreadsheets
Approver groups	HUMAN task assignment policies
LLM instructions	Orkes AI prompt templates
Waves, windows and limits	This file, passed in as workflow input

For the next merger, the FDE coach writes a new version of this file, the rules sheets and the field mappings. The twelve workflows stay as they are.

PLAN

Revised plan for the next three weeks

This week
to Fri 9 Oct

Agree the contract and the merger config file. Build mock data for Bank A and Bank B: inventories, accounts and three integrations.

OUTPUT
Workflow design signed off

Week 1
Mon 12 – Fri 16 Oct

Mock cores and integrations, POC 1 playbooks and governance, and POC 2 end to end with its LangGraph agent.

DEMO
Two inventories in, a reviewed migration plan out.

Week 2
Mon 19 – Fri 23 Oct

poc1_execute_plan with approvals and rollback, POC 5 for credential, routing and transient failures.

DEMO
Wave 1 is cut over and a broken integration repairs itself. A rejected wave is rolled back.

Week 3
Mon 26 – Fri 30 Oct

POC 6 and the verify-and-repair loop, mapping fixes, a full run, then a second merger set up from a new config file.

FINAL DEMO
One merger from assessment to close, then a second one with configuration changes only.

POC 3 and POC 4 are parked. The unified design keeps their slots, so they can be added later without changing these workflows.

Who builds what

Mock banks, adapters, migrate and rollback workflows	**Sai Nivedh**
POC 1 insight-to-action workflow and governance	**Pranav**
POC 1 wave runner (poc1_execute_plan)	**Shreyaas**
POC 2 stack assessment and the LangGraph agent	**Prem**
POC 5 self-heal	**Shayan**
POC 6 reconcile and verify_and_repair	**Roshan**

Decisions

1. **Settled:** we build on the Orkes cluster Bismillah offered, using its OpenAI integration.
2. **Open:** whether each of us can get our own user on that cluster.
3. **Open:** whether we may use an anonymised inventory from a real merger, or stay fully mock.